



Mathieu ASTRUC

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EXPERIENCE

- Airbus** Mar. 2026 – Sep. 2026
Applied AI Engineer - Master Thesis *France*
- Built a hybrid RAG system over 10k+ export licenses and regulatory docs, routing queries between a LangChain/FAISS pipeline and a NL-to-SQL path for structured data ensuring zero generative approximation on field-level queries.
 - Engineered the full pipeline end-to-end: chunking, embedding, reranking, token-cost optimization, evaluation framework, and user feedback loop designed for reliability in a safety-critical environment.
 - Developed AI POCs with cross-functional teams and presented the system to 100+ employees, driving enterprise AI adoption.
- NTNU – Norwegian University of Science and Technology** May 2025 – Aug. 2025
Research Engineer - Intern *Norway*
- Developed an embedded AI interaction stack for a humanoid robot, combining real-time gesture recognition, computer vision and a fine-tuned LLM for domain-specific dialogue.
 - Optimized on-device inference latency through GPU pipeline tuning, enabling real-time responses under hardware and runtime constraints.
 - Implemented a Human-in-the-Loop framework to manage edge cases, improve robustness and support continuous model improvement.
- Banque de France** May 2024 – Aug. 2024
Data Scientist - Intern *France*
- Automated scraping workflows aggregating unstructured public/financial data from multiple sources.
 - Combined OCR, Speech-to-Text and LLMs to digitize raw audio/visual content, feeding a downstream RAG system.
- Comat Specific** Jan. 2025 – May 2025
Machine Learning Engineer – Part-time role alongside studies *France*
- Built a deep-learning OCR pipeline converting legacy hand-drawn 2D engineering sketches into structured machine-readable data, modernizing industrial workflows.

SELECTED PROJECTS

- Publication – HCI International 2026, Montreal (Springer Conference Proceedings)** 2025 – 2026
- Lead author of an accepted HRI paper on an optimized real-time computer vision architecture for gesture recognition, integrating MediaPipe landmarks, lightweight ML classifiers, and low-latency robot actuation.
- Data Science Project – Universidad Politécnica de Madrid** 2025
- **SME Loan Risk Classification:** Won a Kaggle-style ML challenge by building a classifier to accept or deny SME loan applications, optimized with Macro F1-Score. **Keywords:** Classification, Risk Scoring, Macro F1.
 - **Noise-aware Clustering Analysis:** Analyzed a noisy three-variable dataset and identified meaningful clusters through EDA, cluster-number estimation and DBSCAN tuning. **Keywords:** Clustering, DBSCAN, EDA.

EDUCATION

- Esaip – Master of Engineering, Data Science (Grade A - Top 10%)** Sep. 2021 – Sep. 2026
Engineer's Degree (5-year program) - Optimization algorithms, Multicriteria Optimization, ML, Image Processing, Reinforcement Learning, Explainable AI, Probabilistic Modeling... *France*
- Universidad Politécnica de Madrid** Sep. 2025 – Feb. 2026
Study Abroad – Machine Learning, cloud computing, large-scale data architectures *Spain*
- SeAMK – University of Applied Sciences** Jan. 2024 – Jun. 2024
Study Abroad – Machine Learning, Deep Learning, Software Development, Embedded Systems *Finland*

SKILLS

- Programming Languages** Python, C, C++, SQL, Bash
- Data Science & Machine Learning** PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Transformers
- LLM engineering** LangChain, LlamaIndex, Agent workflows, Fine-tuning (LoRA / QLoRA), Vector DBs (FAISS, Chroma)
- Specializations** NLP, RAG, Computer Vision, Multimodal AI, Human Robot Interaction
- Inference & Deployment** vLLM, llama.cpp, Hugging Face TGI, FastAPI, Docker, Kubernetes
- Languages** French (native), English (C1), Spanish (B1)

BEYOND WORK

Former competitive basketball player · Entrepreneurship · Chess · Sports · Research